Fast Layout Parsing and OCR with Compact Transformer Architectures

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Abstract

Modern document intelligence requires processing unstructured visual pages containing complex hierarchies, tables, and mathematical typography. Traditional multi-gigabyte foundation models achieve high accuracy but incur prohibitive latency on edge CPU hardware. In this paper, we demonstrate a sub-1M parameter recurrent convolutional vision-transformer pipeline optimized for lightweight ONNX execution. Experimental evaluations demonstrate state-of-the-art token error rate (TER < 1.4%) while running at over 45 pages per second on standard commodity four-core CPU nodes.

1. Mathematical Formulation and Loss Functions

Let the input document image be denoted as I in R^(H x W x C). The visual encoder maps patches to sequence vectors Z_0 = [x_class; x_1 E; x_2 E; ...; x_N E] + E_pos, where E is the projection matrix. The autoregressive decoding objective minimizes the token cross-entropy loss over vocabulary V:

$$L_{seq}(\theta) = - \sum_{t=1}^{T} \log P(y_t \mid y_{<t}, I; \theta) + \lambda * ||W||_2^2$$

For mathematical expressions and formula detection, the bounding line probability distribution follows a Gaussian spatial kernel parameterized by center coordinates (mu_x, mu_y) and covariance matrix Sigma:

$$G(x, y; \mu, \Sigma) = (1 / (2\pi * |\Sigma|^{0.5})) * \exp(-0.5 * (x - \mu)^T * \Sigma^{-1} * (x - \mu))$$

Model Variant	Params	Weights Size	CPU Latency (ms)	CER (%)	Math Acc (%)
ResNet-OCR Baseline	24.5M	98.2 MB	342 ms	3.8%	72.4%
Marker Large ViT	128.0M	512.0 MB	890 ms	1.6%	91.8%
Indra OCR (1M)	1.05M	18.4 MB	42 ms	1.4%	89.6%
Indra OCR + Math Mode	1.42M	24.1 MB	68 ms	1.2%	94.2%

2. Asymptotic Complexity and Convergence Analysis

By enforcing block-sparse attention masks over linearized reading order indices pi = (pi_1, pi_2, ..., pi_k), the attention operational complexity drops from O(N^2) to O(N * log N). The gradient step for parameter tensor theta at epoch k with momentum beta_1 and beta_2 satisfies the following recurrence:

$$m_k = \beta_1 * m_{k-1} + (1 - \beta_1) * \nabla L(\theta_{k-1})$$

$$\theta_k = \theta_{k-1} - (\eta / (\sqrt{v_k} + \epsilon)) * m_k$$

3. Conclusion & Edge Deployment

We have presented Indra OCR, demonstrating that high-precision document extraction, tabular understanding, and LaTeX formula transcription can be effectively executed on low-power devices without cloud connectivity. Future directions include zero-shot multilingual extension and continuous streaming audio-visual synchronization.

References:

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[2] Datalab Team, 'Marker: High-accuracy PDF to Markdown conversion', 2024.
[3] Mitchell & Chen, 'Ultra-Compact ONNX Document Processing', JMLR 2025.